

Role of Nitrogen in the Pitting Resistance of Cast Duplex CF-Type Stainless Steels[☆]

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ABSTRACT

In this investigation, critical manipulation, observation, and analysis of polarization curves as a function of nitrogen content in cast duplex CF-type stainless steels (SSs) has been performed. These, in conjunction with microstructural observations and atomic absorption spectroscopic analyses of polarization cell solutions, have led to the discovery of nitrogen's major role in the corrosion resistance of SSs. Nitrogen has been found to encourage the retention of chromium in the passive film and near surface layers during corrosion, which improves pitting resistance.

INTRODUCTION

During the past 10 to 15 years, nitrogen has become an important alloy addition to many wrought austenitic stainless steels (SSs). Initially, nitrogen was added as a less costly austenite former than nickel. An additional motivation for adding nitrogen to SSs results from the increased use of low-carbon grades of wrought austenitic SSs in power generation applications, particularly in boiling water reactor (BWR) and pressurized water reactor (PWR) plant piping. The use of low-carbon alloys is necessitated by the need to avoid welding related corrosion problems (i.e., sensitization). Nitrogen has become a common alloying addition to return the strength of low-carbon alloys to that of nominal carbon content alloys. During this material development period, data have accumulated on the effects of nitrogen additions on corrosion properties of wrought SSs, strongly suggesting that nitrogen improves intergranular and pitting corrosion resistance in these steels.^{1,2}

The high-alloy casting industry is adding nitrogen to selected grades of corrosion-resistant castings in response to the demand for higher performance alloys and as a consequence of the success the wrought alloy industry has had with nitrogen additions. Unfortunately, there is a small amount of data on nitrogen effects on the corrosion behavior of high-alloy duplex cast materials. In response, research was initiated to document the magnitude of the improvement in localized corrosion resistance resulting from nitrogen additions in cast duplex SSs and to elucidate the mechanism by which this improvement occurs. This paper describes some of the results regarding the pitting resistance of solution treated CF-

type alloys. CF-type alloys are the cast equivalent of AISI⁽¹⁾ 304 SSs; the major difference is that CF alloys generally contain 5 to 25% ferrite in the microstructure, while AISI 304 SS is nominally fully austenitic.

EXPERIMENTAL PROCEDURE

Alloys and Preparation

Because of the complex interrelationships between ferrite- and austenite-forming elements in duplex SSs (see, for example, the Schoefer diagram),³ it is impossible to vary the concentration of only one component and maintain a constant ferrite content. However, within the range of nickel concentrations commonly found in the alloys of interest, nickel does not appear to be a particularly important variable in the localized corrosion resistance of SSs.^{2,4} Therefore, it was decided to "isolate" nitrogen as a compositional variable by making alloys with fixed Cr, Mo, and C contents and decreasing the nickel content as the amount of nitrogen is increased to maintain approximately constant ferrite.

A base melting stock for CF-type alloys was prepared and cast into cylindrical shapes of a size convenient for remelting in a Balzers⁽²⁾ induction furnace. High-purity Fe, Cr, Ni, Mo, and graphite powders were used to adjust the composition of the base stock to the desired values. This technique enabled careful composition control and alloys of identical residual elemental content to be produced. High-nitrogen ferrochrome facilitated the addition of nitrogen to each melt. The compositions (based on chemical analysis following casting) of the alloys produced in this manner (labeled CF-11 through -14) and two similar alloys (labeled CF-3 and -4) studied in this investigation are given in Table 1. The range of ferrite contents (7 to 15 as measured by a magne-gage) for the alloys produced by the authors (CF-11 through -14) is a result of the inability to precisely predict the efficiency of nitrogen transfer from the ferrochrome to the melt. Based on the results from many test castings, the nitrogen transfer efficiency varies between ~70 and 90%. Alloys CF-13 and -14 were analyzed following casting and were found to contain less nitrogen than the target value; if the intended nitrogen level had been achieved, it is likely that the ferrite content would have fallen in the target range of 6 to 9 (magne-gage).

Cyclic Polarization Pitting Test

The cyclic polarization pitting test (CPPT) is a relatively rapid technique to compare the relative pitting resistance of many alloys

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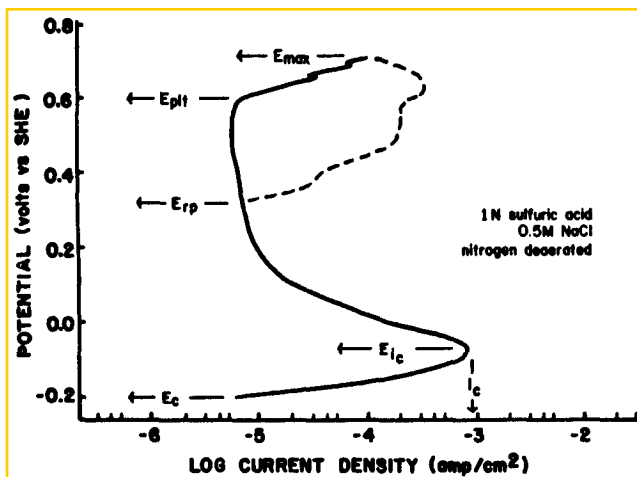


FIGURE 1. Schematic diagram of the CPPT. The characteristic parameters that define the curve are: E_c = open circuit potential; i_c = critical passivation CD; E_p = potential corresponding to i_c ; E_{pit} = pitting potential; E_{max} = potential at which scan direction is reversed ($i = 10^{-4}$ A/cm²); and E_{rep} = repassivation potential.

For this investigation, a test solution of deaerated (nitrogen purge) 1 N sulfuric acid containing 0.5 M NaCl was chosen. The test temperature was 27 °C. Other parameters common to most tests included a 9- μ m diamond paste surface finish and a 3-V/h potential scan rate.

The schematic polarization curve shown in Figure 1 is typical when pitting occurs; shown along the curve are some of the important values characterizing the response of the alloy to the test. The test begins at the open circuit potential (E_c). Once a stable active E_c value was achieved, the sample was polarized anodically at (typically) 3 V/h. The polarization continues through the critical passivation current density (CD) (i_c) through the passive potential range up to the pitting potential (E_{pit}), at which the CD dramatically increases. Polarization is continued to E_{max} , the potential at which pitting has propagated sufficiently to increase the sample CD to 10^{-4} A/cm². The scan direction is then reversed and stopped at the potential where the reverse scan CD equals the upscan CD (E_{rep} —repassivation potential). The magnitude of the CD at which the polarization scan was reversed was chosen to represent a compromise between a value large enough to ensure the onset of stable pitting and one small enough to enable microstructural determination of the majority of pit initiation sites.

During this investigation, it was found that special precautions were necessary to ensure that crevice corrosion at the edges of the sample mount did not occur. Therefore, samples for polarization (Figure 2) were prepassivated in 50% nitric acid (50 °C) following polishing through 9- μ m diamond paste. The edges of the sample were then painted with a tough corrosion-resistant enamel. Immediately before testing, a cotton swab with diamond paste was used to remove the passive film from the exposed portion of the sample, leaving the passive film under the protective enamel at the sample edges. This technique has been shown⁵ to eliminate spurious crevices that form at sample mount edges or at the enamel/sample interface (the latter frequently occurring if the prepassivation step is omitted); otherwise, data interpretation is complicated in at least two ways. Typically, crevice corrosion occurs at much lower potentials than pitting corrosion. If crevice-type corrosion is misinterpreted as pitting corrosion for some alloys or samples and not for others, similar phenomena are not being compared if these mixed corrosion values are used to rank the localized corrosion resistance. In addition, crevices often lead to gradual current increases (from the passive CD) rather than the normal sharp current increase that determines E_{pit} in a standard cyclic polarization test. By using the enamel masking technique, the relative pitting potentials of the exposed sample surfaces may

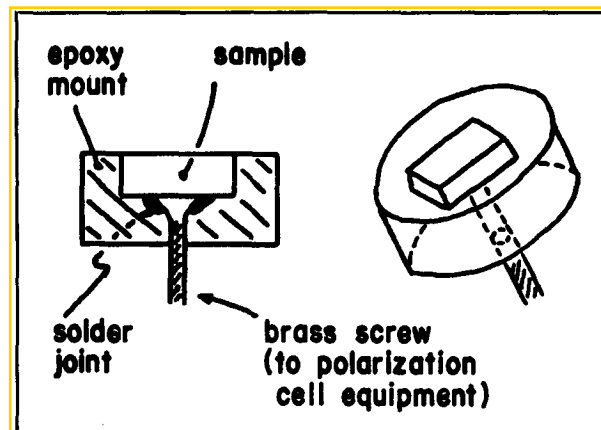


FIGURE 2. Schematic illustration of sample mount for polarization testing.

be compared directly without the need for considering crevice effects that vary from sample to sample.

Atomic Absorption Spectroscopy

The analysis of a polarization curve yields information such as the relative ability of an alloy to passivate or its susceptibility to pitting corrosion. The CD value that is determined at any potential represents the net current at the sample surface and no information concerning the ionic composition of this current is available from the polarization curve. To obtain information on the distribution of ionic species responsible for the anodic current, chemical analysis of the test solution following a polarization experiment must be performed. In this investigation, chemical analyses of test solutions were performed using atomic absorption spectrometry (AAS). A Perkin-Elmer⁽³⁾ Zeeman/3030 system, comprising spectrometer, Zeeman graphite furnace unit, and auto-sampler, was used for all AAS determinations.

Samples were potentiostatically etched in the pitting test solution at potentials near that of the maximum anodic dissolution CD and at a potential just greater than that necessary to induce passivity ($+250$ mV_{SHE}). During dissolution, the CD as a function of time was recorded so that the total amount of charge transferred could be calculated. The concentration of each of the major alloying elements (Fe, Cr, Ni, and Mo) in each test solution, normalized to unit charge transferred, was then determined and related to the bulk composition of the alloy in question.

Determination of the contribution of the major alloying elements to the dissolution current yields important information about the dissolution process as a function of alloy content, but it also indirectly yields information about the surface (passive film and near-surface bulk alloy) composition of the alloy. For example, other factors such as dissolution potential and test environment being equal, a test solution that is determined to have a very low chromium concentration (compared to that following testing of a different alloy) indicates that the corresponding alloy must have a surface relatively rich in chromium. This follows because the chromium must be located in the alloy (unoxidized state), in the passive film (oxidized state), or in the test solution. Such information may be used in much the same way (qualitatively) as information obtained from surface analysis techniques.

RESULTS AND DISCUSSION

To compare the relative pitting resistance of a group of alloys of variable composition, some type of functional relationship that

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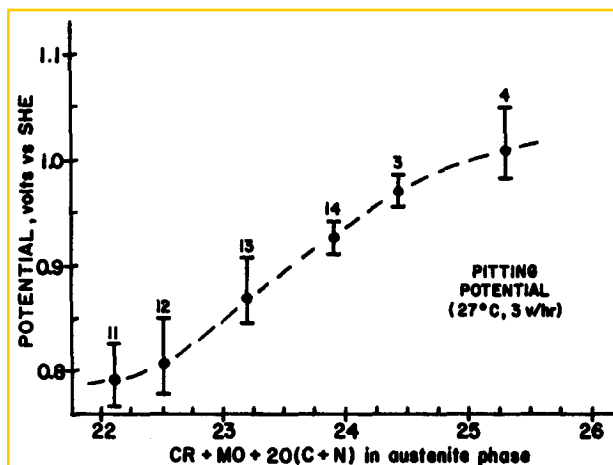


FIGURE 3. Pitting potentials for solution-treated CF alloys determined from the cyclic polarization test. Number above each scatter bar identifies the specific CF alloy.

normalizes the composition of each alloy to a common scale is required. Normalizing expressions have been proposed as a correlative technique by several investigators,^{6,7} and, generally, the expressions contain terms (with various multiplying factors) that consider the bulk Cr, Mo, and N contents. The values that these functions generate are often called "effective chromium contents," or Cr_{eff} . Larger values of Cr_{eff} indicate a more highly alloyed (with elements that promote pitting resistance) material and thus predict a more positive pitting potential.

Using this concept, pitting potential data for solution-treated (1 h, 1120 C, water quench) CF alloys are presented in Figure 3. The normalizing function describing Cr_{eff} is the sum of the weight percentage of Cr and Mo in the austenite plus 20 times that of the carbon and nitrogen in the austenite. Restricting the Cr_{eff} calculation to the composition of the austenite phase is based on the observation that, in every case, pits form exclusively in the austenite phase or at the ferrite-austenite boundaries. In no instance were pits located in the ferrite phase. It should be pointed out that the composition of the austenite phase was not measured analytically but was approximated using partitioning data for Cr between ferrite and austenite in similar alloys⁹ and assuming that all of the nitrogen and carbon in the alloy is in the austenite. Neither approach is exact, but, to a first approximation, they are more appropriate than calculation of Cr_{eff} based upon the bulk composition. The specific form of this Cr_{eff} (including the role of carbon) evolved as a portion of this research² and will not be developed here. For the alloys under consideration (Table 1, Figure 3), nitrogen is the major compositional variable because the Cr and Mo are essentially constant and only CF-3 has a carbon content outside the range of 0.05 to 0.06% carbon.

The pitting potentials plotted in Figure 3 were determined using samples prepared as previously described: specific to these data were a test temperature of 27 C, a potential scan rate of 3 V/h, and a 9- μ m surface finish (including the aforementioned masking procedure) just prior to immersion in the solution of 1 N sulfuric acid and 0.5 M NaCl. (These conditions will be referred to

as the *standard* test conditions.) The data points, all for solution-treated CF alloys, represent the average of at least three determinations, with the vertical bars representing the total scatter in the values. A trend identical to that in Figure 3 was found for a wide range of conditions, such as 9- μ m to 1 grit surface finish, 27 to 42 C solution temperature, and 0.05 to 3.0 V/h scan rate, but this discussion will emphasize the specific results shown Figure 3.

Figure 3 indicates that, as estimated by the CPPT, nitrogen improves the pitting resistance of these solution-treated CF alloys by ~20 mV for each 0.01% N (or ~70 mV per unit Cr_{eff}) addition. Since these solution-treated alloys are essentially carbide free, their pitting behavior cannot be explained entirely by the statement that nitrogen retards carbide formation.⁹⁻¹¹ Likewise, this behavior cannot be rationalized by postulating that increased nitrogen contents result in more favorable partitioning of chromium between the ferrite and austenite phases⁹ because nitrogen improves the pitting resistance of wholly austenitic wrought alloys as well. The suggestion¹² that nitrogen reacts with H^+ to form ammonium ions and locally increase the pH (leading to repassivation) is certainly possible, but this is not likely to have a serious effect in a test solution of bulk pH near 0.5. Apparently, some other mechanism must be operative.

Figures 4 and 5 present typical CPPT curves for the lowest (CF-11) and highest (CF-4) nitrogen alloys. These representative polarization curves exhibit several important features. For example, above the pitting potential, the low nitrogen CF-11 alloy displays a monotonic current increase up to the point (10^{-4} A/cm²) where the scan direction is reversed. However, a common feature of the high nitrogen alloys, represented here by CF-4, is the partial repassivation of pits that appear to be well into the propagation stage. In Figure 5, this is indicated by the two sudden current decreases that occur above 1000 mV. This behavior is unique to the high-nitrogen alloys and will be referred to again.

Figure 6 shows the relationship of the open circuit potential (E_o) and the nitrogen content of the austenite phase. Nitrogen content is plotted rather than the composition normalizing factor (Cr_{eff}) since nitrogen is the major compositional variable. Possible reasons for the decrease in the open circuit potentials as a function of increasing nitrogen content include a change in the cathodic reaction and/or in the anodic (charge transfer) reaction in this environment. The consideration that the amount of ferrite in the alloy could be directly controlling E_o does not really apply to these alloys because CF-13 and -14 have the same amount and slightly more ferrite, respectively, than CF-4, yet the data in Figure 6 follow a smooth curve. A series of very slow scan rate polarization curves was performed for the cathodic and the anodic region of the polarization curve near E_o (effectively a type of polarization resistance experiment). The results indicated that the cathodic reaction in this environment is not changing as a function of nitrogen content, i.e., the exchange CD and slope of the charge transfer region remain constant. Therefore, it is concluded that nitrogen has altered the anodic reaction.

The critical passivation CD (i_c) also increased with increasing nitrogen in the alloy (Figure 7). This contrasts with the notion commonly discussed in the literature¹² that predicts that alloying to lower i_c results in an increase in pitting resistance. In addition to having a value of i_c that increases with nitrogen content, the potential at which this peak occurs becomes more negative as the

TABLE 1
Compositions of Alloys under Consideration

Alloy Designation	Fe	Cr	Ni	Mo	C	N	P	S	Si	Mn	Ferrite No.
CF-11	65.9	20.8	11.7	0.22	0.050	0.025	0.01	0.01	0.7	0.6	7
CF-12	66.9	20.9	10.5	0.22	0.050	0.054	0.01	0.01	0.7	0.7	9
CF-13	68.2	20.8	9.2	0.23	0.050	0.082	0.01	0.01	0.7	0.7	10
CF-14	69.1	21.0	8.0	0.23	0.050	0.098	0.01	0.01	0.8	0.7	15
CF-3	70.3	20.6	7.4	—	0.015	0.170	—	—	1.1	0.4	18
CF-4	70.6	20.5	7.2	—	0.058	0.170	—	—	1.1	0.4	11

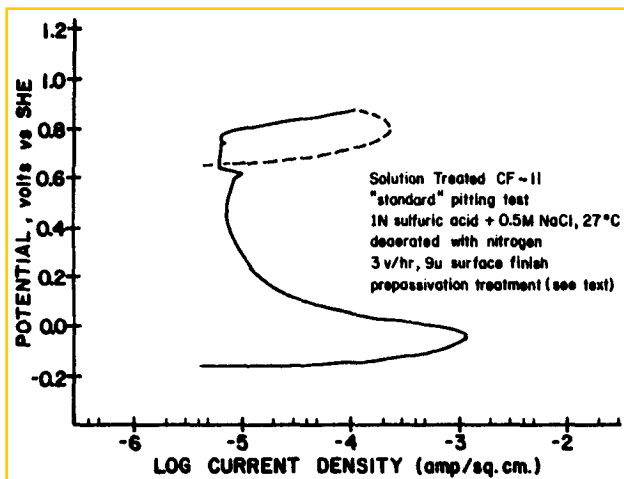


FIGURE 4. CPPT curve for solution-treated CF-11 (low-nitrogen alloy).

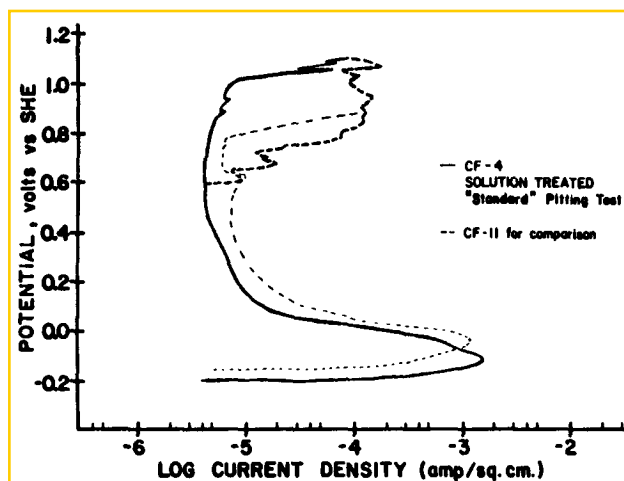


FIGURE 5. CPPT curve for solution-treated CF-4 (high-nitrogen alloy).

nitrogen content is increased. Figure 8 shows the relationship between the active peak potential and nitrogen content. This figure indicates that the potential of the active peak decreased 4 to 5 mV for each 0.01% N increment in austenite. (Similar behavior has been noted for fully austenitic alloys.¹³)

Finally, a progressive change is observed in the shape of the anodic loop (active dissolution prior to passivation) as the nitrogen content is increased. The solid portion of Figure 9 is an expanded potential scale plot of the anodic loop for the highest nitrogen (Cr_{off}) alloy. The unusual shape of the curve (inflection point) near -50 mV suggests an anodic loop consisting of two separate components of the form shown schematically by the dotted curves in Figure 9.

That the active dissolution polarization curve comprises two components is also suggested by the data in Figure 10. This graph compares the relative positions of the active corrosion peaks of CF-11 (low nitrogen) and CF-4 (high nitrogen) with the active corrosion curves of pure iron and pure chromium. It is apparent that the curve for CF-4 is very similar to pure iron during the initial stages of active dissolution. This suggests that nitrogen encourages the active dissolution of iron—and/or retards the dissolution of chromium—and thus enriches the remaining metal surface in chromium. It is, therefore, proposed that the anodic loop for high nitrogen alloys is composed of a low potential loop that represents iron being preferentially dissolved (thus leaving chromium behind) in a magnitude proportional to the nitrogen content and a high potential loop that is representative of all CF alloys

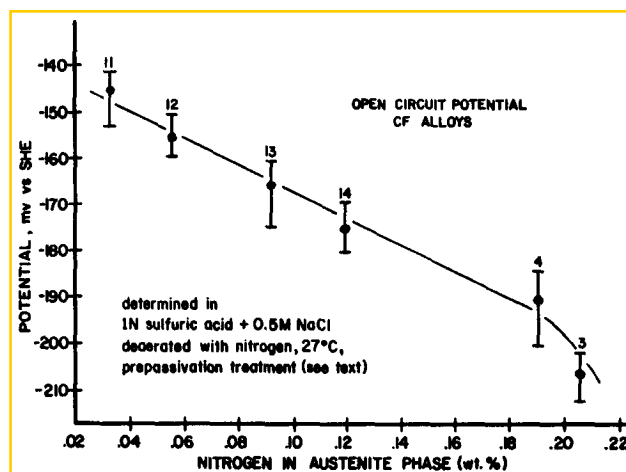


FIGURE 6. Open circuit potentials of solution-treated CF alloys as a function of austenite nitrogen content. Number above each scatter bar identifies the specific CF alloy.

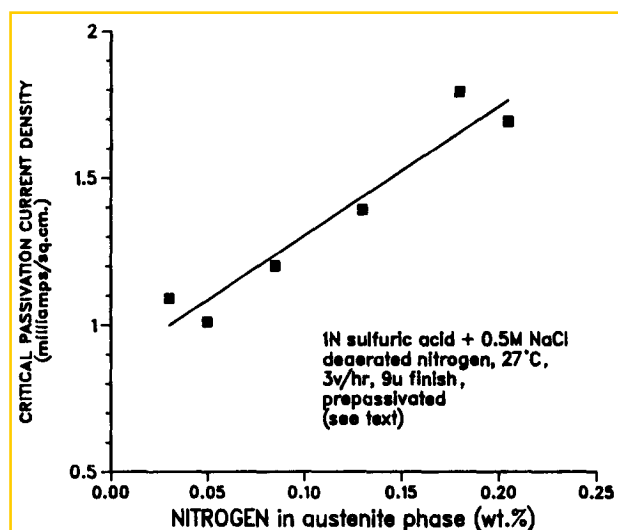


FIGURE 7. Critical passivation CD of solution-treated CF alloys as a function of austenite nitrogen content.

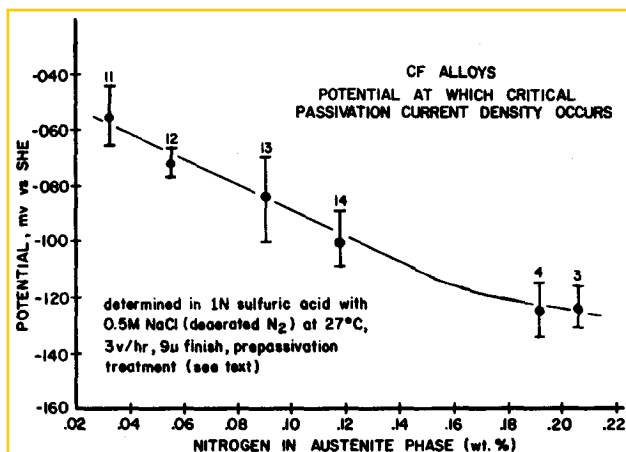


FIGURE 8. Potential at which i_a occurs for solution-treated CF alloys as a function of austenite nitrogen content. Number above each scatter bar identifies the specific CF alloy.

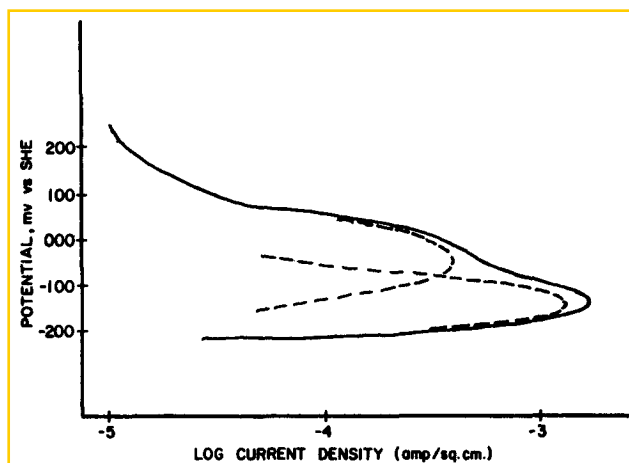


FIGURE 9. Schematic diagram indicating the contribution of two separate anodic loops (dotted curves) to the overall anodic loop (solid curve) for solution-treated CF-4 (high-nitrogen alloy).

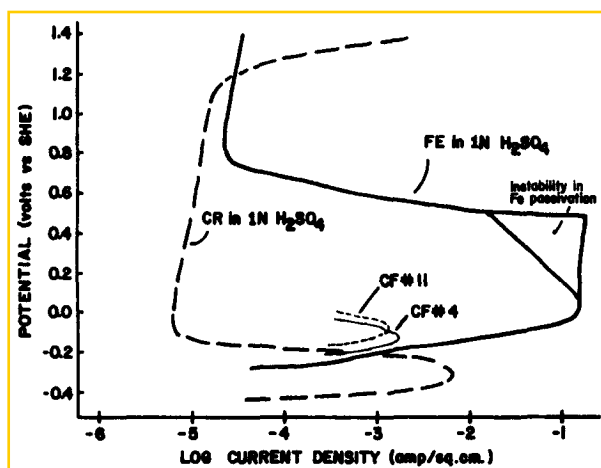
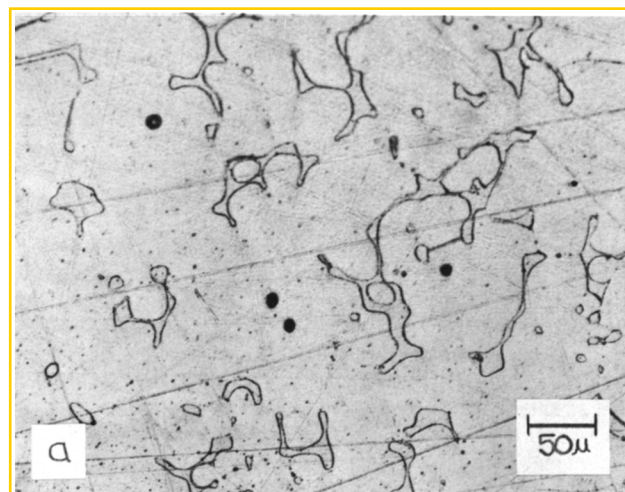
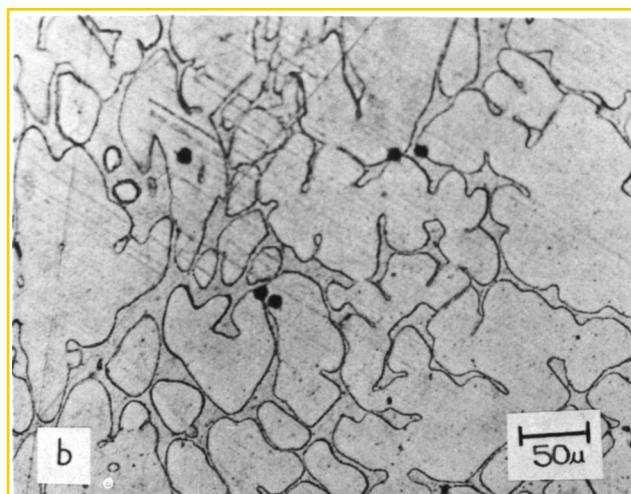


FIGURE 10. Comparison of the relative positions of the active corrosion peaks for CF-11 (low nitrogen) and CF-4 (high nitrogen) with the polarization curves for pure iron and pure chromium.



with common base composition, i.e., composition exclusive of nitrogen.

Although a possible explanation for this phenomenon is that the two "separate" peaks represent dissolution of the austenite and ferrite phases, arguments may be made rejecting this possibility. First, the separation or splitting into two peaks is observed only for alloys of total nitrogen in austenite near or above 0.12% N. Equally important is the fact that the anodic loop splitting is observed in wholly austenitic alloys with nitrogen contents near or above 0.12% N, as shown in this research² and elsewhere.⁹ Anodic loop splitting has also been observed in duplex alloys containing nitrogen, but the authors^{8,14} suggested that the behavior was controlled by the degree of chromium segregation between the ferrite and austenite or that it represented differentiation in the onset of passivation for the two phases. Obviously, neither of these completely explains this phenomenon because of similar results for wholly austenitic alloys.

The location of the pits following a cyclic polarization test also seems to be a function of the nitrogen content (Cr_{eff}) for the alloys referenced in Figure 3. For the relatively low-nitrogen alloys (CF-11 through -13), the vast majority of the pits nucleate and propagate in the austenite phase. However, at the higher nitrogen contents (CF-14, -3, and -4), pit initiation occurs to a large (but not exclusive) extent at the ferrite-austenite boundaries. Figure 11 contains representative microstructures following the standard pit-

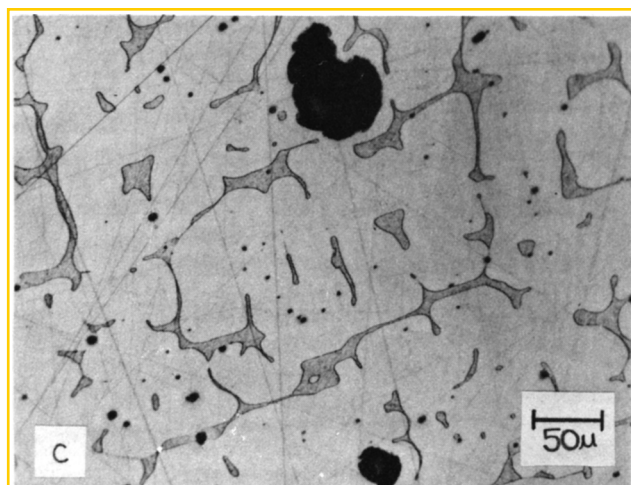


FIGURE 11. Representative microstructures following the CPPT. (a) Pit initiation in CF-11 (low nitrogen) at 200X. Note small pits in austenite. (b) Pit initiation in CF-4 (high nitrogen) at 200X. Note several pits at ferrite-austenite boundaries. (c) Pit propagation in CF-11 at 200X.

ting test that indicate this trend. A conclusion that may be drawn from this observation is that the ferrite-austenite boundaries become susceptible to pitting only after the pitting resistance of the austenite has been improved to some level, ostensibly because of the presence of solid solution nitrogen. This conflicts with the suggestion¹⁵ that nitrogen improves the pitting resistance of duplex alloys due to the elimination of ferrite—and thus ferrite-austenite boundaries—from the microstructure.

In addition to the pit location, the total number of pits seems to be a function of nitrogen content. It is a trend in these alloys that the total number of pits is smaller but the average pit size larger for the high nitrogen alloys. This can be related to the potential at which pitting occurs in these alloys. Nitrogen improves the passive film such that fewer sites are susceptible to pitting and that a higher potential is required to initiate pitting compared to the low-nitrogen alloys; therefore, the high potential at which pitting occurs may cause the pit to grow quite large. This also affects the determination of the repassivation potential (E_{rp}) for these alloys. The fact that the pits in the high-nitrogen alloys are larger (than the low-nitrogen alloys) means that a lower value of E_{rp} is often required to repassivate the pits in high-nitrogen alloys.

Several variations in procedure were used to test the hypothesis that the low potential loop is related to the alloy nitrogen content and that the dissolution that occurs at this potential influences the pitting resistance. The first variation was to perform pitting experiments on the high-nitrogen alloys in which the potential was changed in such a manner that allowed the lower potential loop to be omitted, or at least have its influence minimized. This was done by performing exactly the same cyclic polarization procedure as the previously described test, with the exception that the potential region from a few millivolts below E_c up to a potential of -80 mV (corresponding to a potential near, but just below, the peak of the high potential loop of Figure 9) was scanned very quickly (>200 V/h). At this point, the potential scan was allowed to proceed normally. This procedure of omitting the potential range from near E_c to -80 mV was performed on the CF-11 (low nitrogen) alloy as well. The results, shown in Figure 12, indicate that this procedure has no effect on the CF-11 alloy (because no "important" portion of the anodic loop was omitted) but that the CF-3 and -4 alloys show a significant drop in the value of E_{pit} when the lower loop is omitted in the upscan. Thus, some factor relating to the higher corrosion resistance of the high-nitrogen alloys is influential in the potential region that was omitted.

The polarization curve in Figure 13 is representative of this evaluation procedure. It indicates that while the net current was still cathodic, the potential was rapidly moved from just below E_c to -80 mV in ~ 2 s. In the high-nitrogen alloys (CF-3 and -4), upon reaching -80 mV, a large anodic current surge of 2 to 3 s occurred. In the low-nitrogen alloy (CF-11), this current surge did not occur. In addition to having an E_{pit} considerably below the E_{pit} predicted by the standard procedure, the high-nitrogen alloys also exhibited uncharacteristic large current excursions at potentials below E_{pit} that are absent during the standard procedure. Also significant is the fact that, by virtue of omitting the low potential loop for the high-nitrogen alloys, the pit initiation sites changed from predominantly ferrite-austenite boundary locations to exclusively within the austenite phase. As shown earlier, the austenite more readily pits (as measured by E_{pit}) unless the austenite is somehow made more pit resistant such that the ferrite-austenite interface is more susceptible to pitting as the potential is increased. Thus, mechanisms exist to enhance the pitting resistance of the passive film on the austenite if the potential scan includes the low potential loop. It appears that the current surge that occurs for the high-nitrogen alloys at -80 mV is sensitive to the nitrogen content and hence to the beneficial retention of chromium in the passive film. If this were not the case, the pitting potential of the high-nitrogen alloys would be identical to that for the low-nitrogen CF-11 alloy for which this current surge does not occur. Its absence for the CF-11 alloy also implies that dissolution is encouraged by nitrogen during the current surge for the high-nitrogen alloys. The influence

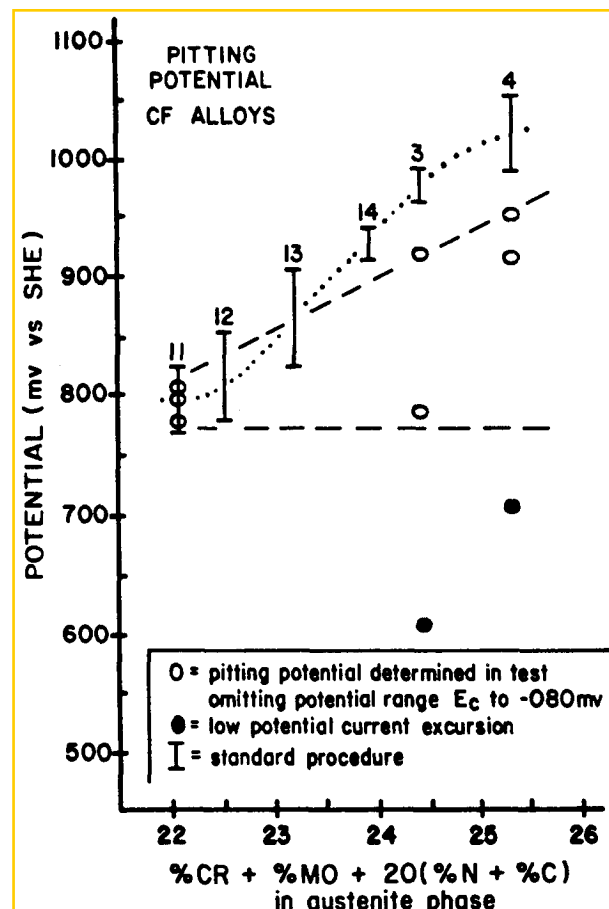


FIGURE 12. Comparison of pitting potentials determined from the standard cyclic polarization test with pitting potentials determined from testing which limits the influence of the low potential loop in the scan procedure.

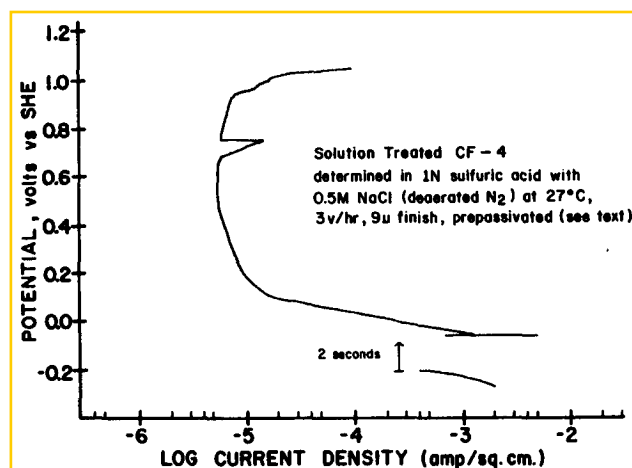


FIGURE 13. Representative polarization curve for test procedure in which the potential region corresponding to the low potential loop was omitted by very rapid scanning (~ 200 V/h).

of scanning through the potential range of the lower loop is further demonstrated by the following sequence. Following omission of the anodic loop below -80 mV for the high-nitrogen alloys, the potential was scanned back down into the low potential loop region prior to completion of the pitting test. On subsequent scanning to the pitting potential range, E_{pit} values essentially equal to those resulting from a continuous scan from E_c are observed for the high-nitrogen alloys.

Major evidence that the low potential loop represents the enrichment of chromium in the surface layers of the alloy was obtained by analyses of the corrosion products in the test solution following various dissolution procedures at potentials near the anodic loop(s). The analysis of the corrosion products was performed using atomic absorption spectroscopy (AAS). This technique, using high-temperature graphite furnace technology to vaporize atom species, enabled reproducible measurements of Fe, Cr, Ni, and Mo concentrations in the parts per billion (ppb) range to be performed on the polarization cell aqueous environment.

Samples prepared in the standard way were potentiostatically held at the selected potential (or rapidly scanned to this potential) for sufficient time to collect a measurable amount of corrosion products in the 1.8-L test solution. During dissolution, the current was continually recorded as a function of time so that the total dissolution charge could be calculated. The concentrations of Fe, Cr, Ni, and Mo in the solution were then determined and normalized to unit dissolution charge. These concentrations were then converted to a percentage of the corrosion product (assuming the contribution from all other elements is negligible). The percentage of the corrosion product assigned to each element may then be related to the percentage of the respective element in the bulk alloy. When the ratio (percentage in corrosion products divided by percentage in the bulk alloy) is less than one, the particular element is not going into solution in amounts as great as its presence in the bulk alloy and is therefore being enriched on the sample surface and/or in the passive film.

The total amount of corrosion product measured for the active dissolution experiments is typically 10 or 15% greater than would be expected compared to the total number of coulombs passed during the experiment. The magnitude of this discrepancy is consistent from test to test and its source is unknown; two sources are the corrosion that occurs during the wait (3 min) for a stable E_c before starting the test and the assumption of the valence of the discharged ion. For the passive potential experiments the error between the measured corrosion product and the known number of coulombs passed during the test was somewhat larger than for the active corrosion experiments. Because of the low net anodic currents involved in the passive range, corrosion products accumulating during open circuit corrosion would have an even larger effect on the total corrosion products for a passive potential test. Errors of this type are essentially constant from test to test; thus, when comparing relative percentages, the importance of the absolute magnitude of the corrosion product concentration is diminished.

Table 2 contains the AAS results for potentiostatic dissolution in the 1 N sulfuric acid and 0.5 M NaCl environment. The dissolution potentials were near the peak critical passivation CD, near the second "peak," if one exists, and just into the passive potential region at +250 mV. Results from two additional alloys, CF-1 and -2, have been included in this table. These alloys (CF-1 = 17% Cr, 0.025% C, 0.025% N; and CF-2 = 17% Cr, 0.058% C, 0.025% N) are very similar to the alloys listed in Table 1 except for the low Cr content. They were included to expand the range of Cr_{eff} in the alloys analyzed by AAS. Since chromium is the major element controlling passive film behavior, the AAS results are

summarized graphically in terms of the ratio defined by the fraction of chromium found in the corrosion products divided by the fraction of chromium in the bulk alloy in Figure 14. The data indicate that as the Cr_{eff} increases (effectively nitrogen content), the amount of Cr in the test solution decreases. For dissolution near the potential of the critical passivation CD (active peak), the decrease is significant. The change in the $Cr_{corrosion\ products}/Cr_{bulk\ alloy}$ ratio with Cr_{eff} correlates to the change in E_{pit} as a function of Cr_{eff} . As the value of this ratio decreases, the amount of Cr in the alloy surface increases and the pitting potential should then increase. This is indeed the case.

For the high-nitrogen alloys, dissolution at the potential of the higher anodic loop (second peak) produces a much higher amount of chromium in the test solution (compared to the lower anodic peak potential). Consistent with the pitting behavior of samples for which the upscan omitted the lower potential loop, the pitting resistance of the alloys so treated is also much lower.

The Figure 14 data indicate that increasing nitrogen also increases retention of chromium during passive state dissolution but that the retention is not as strong as for active state dissolution. It would appear that in addition to improvement of the passive film due to the retention of chromium, nitrogen aids the maintenance of passivity in the following way. Assuming that an active pit (or pit embryo) corrodes similarly to the active corrosion of the alloy as a whole, the current burst in a high-nitrogen alloy pit embryo contains less chromium than the current burst in a low-nitrogen alloy. The resulting high-chromium material left behind is more capable of repassivating than the original "starting material." For alloys that do not retain chromium to this degree during active dissolution (i.e., pitting), this is not true. Therefore, this predicts that a passivation procedure involving a particular anodic loop is not necessary for nitrogen to be beneficial to pitting resistance because (1) nitrogen also retains Cr during passive-state dissolution (although to a lesser degree than active dissolution) and (2) nitrogen encourages the repassivation process immediately following pit nucleation.

In an attempt to confirm this concept, a set of cyclic polarization experiments were performed in which the magnitude of the anodic loop (active dissolution prior to passivation) was significantly diminished. This was accomplished by first allowing the anodic loop portion of the polarization curve (from E_c to 200 mV) to occur in 1 N sulfuric acid that was free from chlorides. Sufficient NaCl was then added with a separation funnel through a port in the top of the test cell to bring the chloride concentration of the entire solution to 0.5 M at the potential of 200 mV. Pitting potentials determined from the subsequent upscan are shown in Figure 15. A large difference is not observed between the values of E_{pit} obtained with the standard test procedure (relatively large anodic loop) and the procedure described here. The results in Figure 15, based on scanning the anodic loop in the absence of chloride, are not directly comparable to results from the standard pitting test because the conditions of passive film formation are different. In the former, chloride ions are not incorporated into the passive film during its formation, and consequently, slightly higher E_{pit} values might be expected to result, particularly for the low-nitrogen alloys where resistance to pitting is more variable. However, the relative

TABLE 2
AAS Results for Potentiostatic Dissolution in 1 N H_2SO_4 and 0.5 M NaCl

Alloy	Cr_{eff}	E_{pit}	Active Peak			Passive (250 mV)			Active, Second Peak		
			Fe	Cr	Ni	Fe	Cr	Ni	Fe	Cr	Ni
CF-1	17.5	510	0.95	1.59	0.37	0.79	1.36	2.49			
CF-2	18.5	710	0.92	1.22	1.55	0.84	1.15	2.57			
CF-11	22.1	795	1.14	0.78	0.73						
CF-12	22.5	810	1.17	0.66	0.46						
CF-13	23.2	860	1.26	0.41	0.58						
CF-14	23.9	920	1.15	0.49	1.29						
CF-3	24.4	960				0.96	0.79	2.20	0.89	1.22	1.66
CF-4	25.3	1010	1.08	0.40	2.17	1.00	0.72	2.06	0.95	1.09	1.46

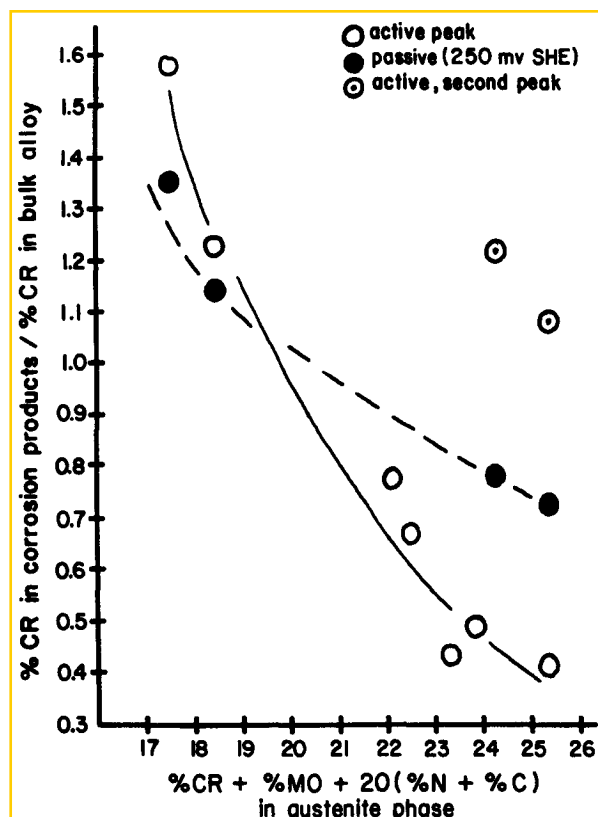


FIGURE 14. Accumulation of chromium in the passive film and near surface layers as a function of Cr_{eff} for solution-treated CF alloys.

pitting potentials among alloys exposed to a particular evaluation procedure may be directly compared.

As intended, the maximum anodic currents for the samples polarized in chloride-free sulfuric acid are all about a factor of 30 lower than for the equivalent alloy in the standard solution ($\sim 4.5 \times 10^{-5}$ vs 1.5×10^{-3} A/cm², respectively). Despite the lower anodic currents, the Figure 15 data nevertheless indicate some benefit of solid solution nitrogen in terms of pitting resistance. This suggests that whenever nitrogen is present in the alloy, independent of the actual size of the anodic loop, chromium tends to be retained in the passive film. Figure 16 indicates that the open circuit potential of the alloys in the chloride-free sulfuric acid varies as a function of nitrogen in a very similar fashion to the E_p values for the standard test solution, although the values in the presence of chloride are ~ 140 mV lower. Chlorides are known to increase the anodic CD in the active potential range more than the CD for the cathodic reaction; hence, the decrease in E_p in the presence of chlorides is to be expected. The general conclusion, however, is that nitrogen may exert an influence on the active dissolution behavior for a wide range of environments.

In view of the preceding discussion, it seems plausible to suggest that the role of nitrogen is related to chromium enrichment at/near the surface in these alloys. On an atomic scale, it is not known by what mechanism this occurs. It can be speculated that because of the known affinity of chromium for nitrogen, the fact that nitrogen slows chromium diffusion in the austenite^{16,17} and that nitrogen accumulates at the metal/oxide interface during corrosion,¹⁸ chromium is retained in the passive film (and near surface layers) because of a chromium/nitrogen interaction (similar to a clustering effect), which inhibits both diffusion and dissolution. This specific mechanism, however, has not been verified.

CONCLUSIONS

The present research, through critical manipulation, observation, and analysis of cyclic polarization curves as a function of ni-

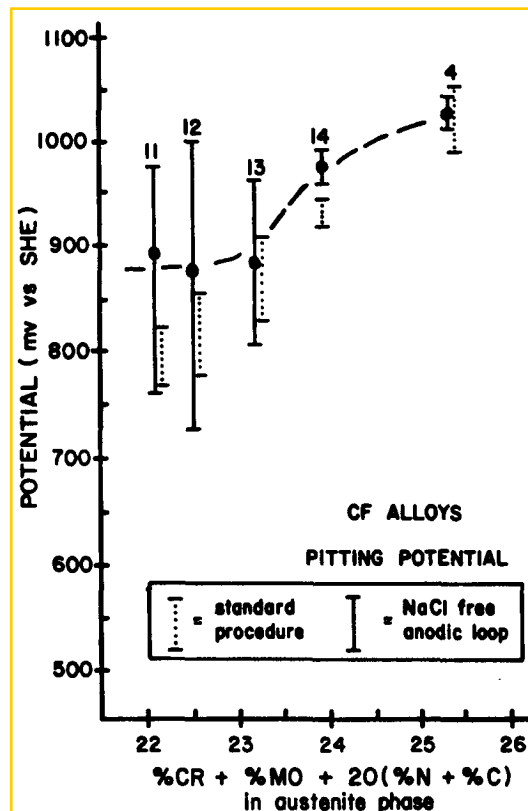


FIGURE 15. Pitting potentials of solution-treated CF alloys determined by cyclic polarization procedure modified such that no NaCl was present in solution during active dissolution.

trogen content of the CF alloys, has led to a description of a mechanism of the improvement in pitting resistance resulting from nitrogen additions. It is hypothesized that the change in shape and position of the anodic loop as a function of nitrogen content results from contributions from two dissolution processes that occur in a magnitude related to the amount of nitrogen in austenite. Specifically, for high-nitrogen contents, the anodic loop of the cyclic polarization curve splits into two active loops. The high potential loop was found to be common to all alloys of this general composition class, but the low potential loop increased in magnitude and occurred at lower potentials as the nitrogen content increased. The observation that this low potential loop closely resembles the polarization curve for the active dissolution of iron in similar solutions suggests that nitrogen encourages the dissolution of iron and the accumulation of chromium during the initial stages of active dissolution.

The pitting of high-nitrogen alloys was observed to be very sensitive to procedure variations designed to vary the influence of the active dissolution processes occurring on passing through the low potential loop (i.e., during the early stages of active dissolution). The pitting test results for the low-nitrogen alloys were not influenced by these procedure variations.

It was also shown, with atomic absorption spectroscopy (AAS), that increasing nitrogen in CF alloys decreases the amount of chromium found in solution following potentiostatic dissolution at a variety of potentials. Since the chromium content of the solution is lower with increasing nitrogen content of the alloy, chromium must be accumulating at the surface (in the passive film or near surface layers) of the nitrogen-containing alloys. Thus, in the presence of solid solution nitrogen, the surface chromium content of the alloy is effectively increased and the pitting resistance is observed to improve. The observed trend of the nitrogen effect on chromium accumulation was observed at all dissolution potentials examined, but this effect was greatest for active dissolution at potentials near that of the low potential loop.

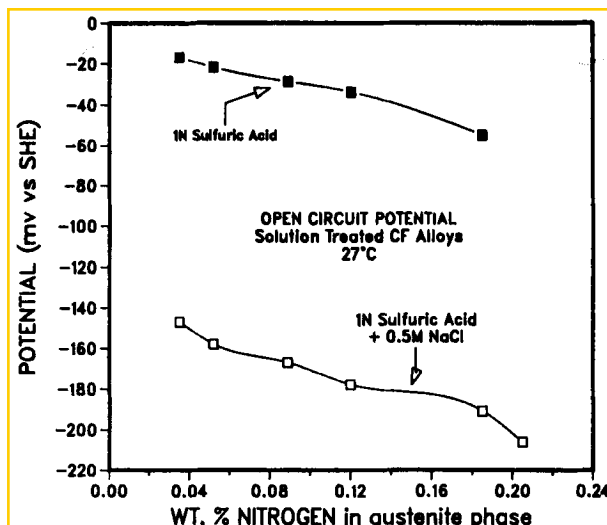


FIGURE 16. Comparison of E_c values in 1 N sulfuric acid with and without 0.5 M NaCl for solution-treated CF alloys

It is suggested that the completion of an anodic loop potential scan is not necessary to develop pitting resistance resulting from nitrogen additions. During the initial burst of current associated with a pit embryo, active dissolution conditions exist. In the presence of nitrogen, this current burst contains predominantly iron ions and the pit surface remaining becomes relatively enriched in chromium and thus may repassivate more easily than in the absence of nitrogen. This was observed experimentally in the cyclic polarization curves for CF alloys. In the passive potential region (below the pitting potential), the high-nitrogen alloys exhibit much smaller (and fewer) current excursions than the low-nitrogen alloys prior to the onset of stable pitting. In addition, the high-nitrogen alloys were observed to exhibit some tendency toward partial repassivation at potentials above that determined to be the pitting potential.

Finally, it is proposed that, on an atomic scale, the affinity of nitrogen for chromium discourages chromium dissolution resulting

from a clustering type of effect. Although the clustering-type mechanism was not analytically proven, this concept is consistent with other observations noted in the literature, such as the facts that nitrogen retards chromium diffusion in the austenite and that nitrogen accumulates at the metal/oxide interface of SSs during corrosion.

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